Force

In physics, a **force** is a push or pull that acts on an object as a result of its interaction with another object.

It is an external agent that can alter an object’s state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity** – Forces have both magnitude (strength) and direction.

- **Unit** – The standard metric unit for force is the **Newton (N)**.

- *One Newton* is the force required to accelerate a 1 kg mass at a rate of **1 m s^{-2}**.

- **Formula** – Newton’s Second Law of Motion:

$$
\mathbf{F} = m\,\mathbf{a}
$$

where **F** is force, **m** is mass, and **a** is acceleration.

Common Types of Forces

1. **Contact Forces**

- Act through physical contact between objects.

- Examples: friction, tension, air resistance, and a physical push or pull.

2. **Non-Contact (or Field) Forces**

- Act over a distance without physical contact.

- Examples: gravity, electrostatic forces, magnetic forces, and nuclear forces.

> **Note:**

> - Forces are always represented as vectors, so both magnitude and direction are essential.

> - The Newton (N) is defined as the force that gives a 1 kg mass an acceleration of 1 m s■^{2}.

> - Newton’s Second Law provides a direct relationship between force, mass, and acceleration, making it a fundamental tool for analyzing motion.